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Symbiosis Institute of Technology, Pune

Faculty of Engineering

CSE - Academic Year 2026-27

Java Programming – Lab (Batch 2025-29)

Lab Assignment No:- 19

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Batch	2025-29
Class	Programming with JAVA
Academic Year & Semester	2nd Year – 3rd Sem
Title of Assignment	Develop a Java application to establish connectivity with a database and perform data retrieval using the SELECT operation.
Practice questions	1. Create a Java application to display student records from database using SELECT query.

```
import java.sql.*;

public class StudentRecordsDemo {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/testdb";
        String user = "root";
        String password = "XXXXXXX";

        String query = "SELECT roll_no, name, course FROM students";

        try (Connection conn = DriverManager.getConnection(url,
user, password);
             Statement stmt = conn.createStatement();
             ResultSet rs = stmt.executeQuery(query)) {

            System.out.println("---- Student Records----- ");
            while (rs.next()) {
                System.out.println("Roll No: " +
rs.getInt("roll_no")
                        + " | Name: " + rs.getString("name")
                        + " | Course: " + rs.getString("course"));
            }
```

```
        } catch (SQLException e) {
            System.out.println("Database error: " + e.getMessage());
        }
    }
}
```

2. Create a Java application that connects to a database and displays product details using SELECT query. Show product ID, product name, quantity, and price.

```
import java.sql.*;

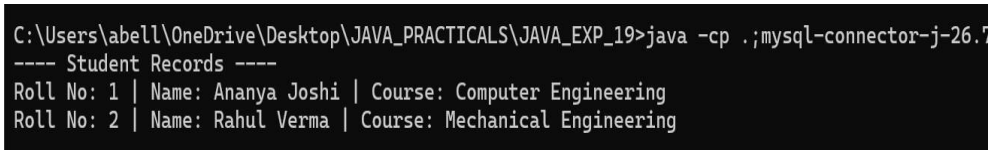
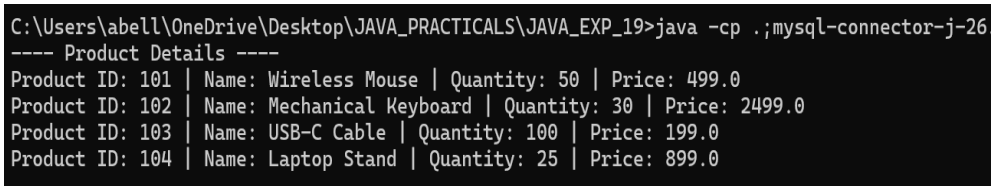
public class ProductDetailsDemo {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/testdb";
        String user = "root";
        String password = "XXXXXXX";

        String query = "SELECT product_id, product_name, quantity,
price FROM products";

        try (Connection conn = DriverManager.getConnection(url,
user, password);
             Statement stmt = conn.createStatement();
             ResultSet rs = stmt.executeQuery(query)) {

            System.out.println("---- Product Details----- ");
            while (rs.next()) {
                System.out.println("Product ID: " +
rs.getInt("product_id")
                        + " | Name: " + rs.getString("product_name")
                        + " | Quantity: " + rs.getInt("quantity")
                        + " | Price: " + rs.getDouble("price"));
            }

        } catch (SQLException e) {
            System.out.println("Database error: " + e.getMessage());
        }
    }
}
```

Output	1)  2) 
Conclusion	JDBC provides a standard, driver-based way for Java applications to communicate with relational databases without depending on a specific vendor's API. Using DriverManager to open a Connection, a Statement to run SQL, and a ResultSet to read the returned rows, a Java program can retrieve and process live database records — the foundation for most real-world desktop, web, and enterprise data access.